Homework #3

Problem 1: Please calculate the one-loop renormalization group equation for the theory

$$\mathcal{L} = \partial_\mu \phi \partial_\mu \phi^* - m^2 |\phi|^2 - \frac{\lambda}{2}(|\phi|^2)^2 \ , \tag{1}$$

where ϕ is a complex scalar.

Problem 2: Please first derive all the counter terms using dimension regularization and the minimal subtraction condition for QED and then derive the one-loop renormalization group equation using the BPHZ method.

Problem 3: Please use dimensional regularization and minimal subtraction method to calculate the one-loop β function for a SU(N) Yang-Mills theory with N_F fermions in the fundamental representation.

Problem 4: The Lagrangian for a complex scalar can be written as

$$\mathcal{L} = \partial_\mu \phi \partial^\mu \phi^* + m^2 |\phi|^2 - \frac{\lambda}{2}(|\phi|^2)^2 \ . \tag{2}$$

In the class, we presented two ways to decompose ϕ. The first one is

$$\mathrm{Re}\phi = v + \frac{1}{\sqrt{2}}\phi_1 \ , \ \ \mathrm{Im}\phi = \frac{1}{\sqrt{2}}\phi_2 \ . \tag{3}$$

The other one is

$$\phi = (v + \frac{1}{\sqrt{2}}R) \exp\left[\frac{i\theta}{\sqrt{2}v}\right] \ . \tag{4}$$

In the first decomposition, the goldstone is ϕ_2, while in the second one, the goldstone is θ. Please calculate the tree-level four-point scattering matrix element of the goldstone boson for the both the two decomposition, and show that they are indeed the same.